VENKATRAMAN R

Data Scientist | AI/ML Engineer | Business & Data Analyst
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LinkedIn | GitHub | Portfolio

SUMMARY

Entry-level Data Scientist who builds machine learning models for **fraud detection** and sales forecasting. Built **7 end-to-end projects**, including a fraud model with **0.97 AUC-ROC**, and put several live on Streamlit. Strong in **Python, SQL and Power BI**. 1.3 years of work experience handling operations data and reports. Looking for entry-level Data Science, ML and Data Analyst roles in **Chennai and Bengaluru**.

SKILLS

Programming & Data: Python, SQL, PostgreSQL, Pandas, NumPy
Machine Learning: Scikit-learn, XGBoost, Random Forest, SMOTE, Optuna, SHAP, Feature Engineering
Deep Learning & GenAI: TensorFlow/Keras, LSTM, LangChain, RAG, FAISS, Prompt Engineering
BI & Visualization: Power BI (DAX, Power Query), Excel, Plotly
Tools: Streamlit, Docker, Git/GitHub, GitHub Actions, SAP

PROJECTS

Multimodal Quality Inspection System (AI/ML Project)

Tools: Python, Scikit-learn, Pandas, NumPy, Computer Vision, Signal Processing

- Built a system that checks industrial parts using **sound and images** and labels each part OK or NG (not good).
- Combined audio features and image features into one decision (multimodal fusion).
- **Result: 95% validation accuracy, 0.8025 ROC-AUC.** Scored 100 unseen samples and built quality alert reports for each station.

AskMyData: GenAI Data-Analysis Agent — GitHub | Live Demo

Tools: Python, LangChain, FAISS, RAG, Docker, Plotly, Streamlit

- Built an AI assistant that answers plain-English questions about data by writing **SQL and Pandas** code and showing charts.
- Used RAG with FAISS search to reduce wrong answers (hallucination); added a Docker sandbox, input checks and chat memory.
- **Live app** deployed on Streamlit Cloud.

FraudGuard: Mobile Transaction Fraud Detection (Capstone Project) — GitHub

Tools: Python, XGBoost, SMOTE, Optuna, SHAP, Streamlit

- Built a model that spots fraud in mobile transactions. **Result: 0.97 AUC-ROC, 0.92 F1.**
- Fixed imbalanced data with SMOTE and tuned the model with Optuna.
- Used SHAP to explain why each transaction was flagged; live Streamlit app gives real-time risk scores.

BankShield: Loan Fraud Detection | *Random Forest, SHAP, Streamlit* — GitHub | Live Demo

- Built a loan fraud model with **0.94 precision and 0.89 recall**; deployed as a live app.

SalesPulse Forecaster: Sales Forecasting | *TensorFlow/Keras (LSTM), PostgreSQL* — GitHub

- Built an LSTM model that forecasts the next 12 months of sales (MAPE 27.66%).

SalesPulse: Power BI Sales Dashboard | *Power BI, DAX, Star Schema* — GitHub

- Built a dashboard on **$12.6M+** of sales data with **44 custom DAX measures** (customer lifetime value, year-over-year margin).

DataHunt: ETL Framework | *Python, Pandas, openpyxl* — GitHub

- Built a reusable data pipeline that cleans **51,000+ records** and saves them as Parquet and Excel files.

EXPERIENCE

Service Engineer (Operations & Analytics), TVS Vehicle Mobility Solutions Private Limited — Sep 2024 – Dec 2025

- Managed service work for **400 to 500 commercial vehicles per month** (about **INR 4 Cr** monthly revenue).
- Tracked jobs, customers and service records in **SAP**; prepared **Excel MIS reports** for management.
- Assigned jobs to technicians, tracked turnaround-time (TAT) targets, and worked with the technical team on fault diagnosis.

EDUCATION

Professional Graduate Certification in Data Science & AI, Boston Institute of Analytics — Jan 2026 – Jun 2026
B.E. Automobile Engineering, VISTAS, Chennai (CGPA 7.25) — 2021 – 2024

CERTIFICATIONS

IBM SkillsBuild: Artificial Intelligence Fundamentals — Oct 2026
Deloitte Data Analytics Job Simulation, Forage — Sep 2026